ends in a minute air-cell. The tissue of the lungs is composed of fine areolar membrane, air-cells, bronchial tubing, arteries, veins, and capillaries. The air-cells are composed of microscopic sacs of membrane. The bronchial tubes, veins, arteries, and capillaries of the lungs are all attenuated to the utmost degree of fineness, and so profusely distributed that the entire mass of the lungs may be said to consist of an immense surface of membrane, with blood-vessels on one side and air-cells on the other. The blood and air do not come into contact, although the fine and permeable character of the membrane permits the nutritive quality of the blood to pass into and nourish the air-cells, and the vitalizing properties of the inhaled air to act upon and change materially the character of the blood. It has been estimated that there are at least six hundred millions of air-cells in the lungs, each one of which is supplied by an artery communicating with a vein, and inflated by the air inhaled in the act of inspiration. The surface presented by the membrane of the lungs would, it is supposed, cover more than thirty times that of the whole body, and the capacity for inflating this light, spongy tissue is provided for, first, by the elastic intercostal muscles which connect the ribs together and expand with every respiration, and next by the action of the large muscular membrane called the diaphragm, which separates the thorax from the abdomen, and alternately curves upward and downward with the actions of inspiration and exhalation performed by the lungs. The chief function of the respiratory organs is that of aerating the blood,